

Repair Request - Unified State Transition

Concurrency / Performance Review - Canonical States Unchanged

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Supersedes	RR-STS-001 v1.5
Revision Scope	Map transition correctness to concurrency and worker performance verification without adding/changing states.
Approval	Business Owner / Project Manager + Architecture/QA review - pending

Revision rule / กติกาการแก้ไข: เอกสารฉบับนี้เป็นการปิด Audit ด้าน Performance/Technical Design เท่านั้น ไม่เปลี่ยน State, Role, Guard, SLA, Permission, Notification Recipient หรือ Business Semantics ที่ล็อกไว้เดิม หากคำตัดสินด้าน Performance ใดไม่มีอยู่ใน Business Baseline จะระบุเป็น **Gap / Proposed Technical Decision / Pending Approval** และไม่ถือเป็น Requirement เดิมโดยอัตโนมัติ.

Document control note: หน้านี้เป็น Document Control ที่มีผลสำหรับ revision นี้และ supersede ค่า Version/Status บนหน้าปกของ source baseline ที่ถูกรักษาไว้ในหน้าถัดไปเพื่อ traceability.

วงจรสถานะระบบ / STATE TRANSITION

Repair Request System - Unified State Transition

Document Control	Value
Document ID	RR-STS-001
Version	1.5 - Baseline Revision Candidate
Status	Content Complete - Ready for Baseline Approval
Revision Date	12 September 2026
Approval	Business Owner / Project Manager - formal name/signature pending
Sources	RR-REQ-001 v1.5; BR-RR-BASELINE v1.5
Audience	Business Owner, SA, Developer, Tester, Project Manager
Language	ภาษาไทยเป็นหลัก พร้อมชื่อ Entity / Role / State / API เป็นภาษาอังกฤษ
Purpose	กำหนด canonical states, transition IDs, guards, side effects, SLA events and invalid transitions.

Revision rule: เอกสารฉบับนี้ปิด Audit findings ของชุดวันที่ 11 September 2026 โดยแยกสิ่งที่เป็น Business Baseline ออกจาก Technical Design อย่างชัดเจน. การลงนามอนุมัติจริงยังเป็นหน้าที่ของ Business Owner / Project Manager.

1. State Definitions & Global Rules

UI/API must invoke allowed actions; no direct state update or skip. Every transition validates identity, role, tenant/site scope, ownership/assignment, current state, guard, required data and rowversion. Significant actions write append-only audit and correlation ID.

Object	Allowed states / codes
Repair Request	DRAFT, SUBMITTED, UNDER_REVIEW, APPROVED, REJECTED, CANCELLED, CONVERTED
Work Order	OPEN, SCHEDULED, IN_PROGRESS, AWAITING_SUPERVISOR_REVIEW, AWAITING_CUSTOMER_ACCEPTANCE, COMPLETED, CORRECTIVE_ACTION_REQUIRED, CORRECTIVE_PLAN_PENDING, CORRECTIVE_PLAN_APPROVED, CLOSED, CANCELLED
Service Visit	SCHEDULED, RESCHEDULED, IN_PROGRESS, COMPLETED, MISSED, CANCELLED
Work Session	CHECKED_IN, PAUSED, CHECKED_OUT
Corrective Action	DRAFT, PENDING_PLAN_APPROVAL, APPROVED
Time Correction	PENDING, APPROVED, REJECTED

2. Repair Request Transition Matrix

ID	From	Action	Actor	Guard / Required Data	To	Side Effect / Audit
ST-RR-001	DRAFT	Create/Edit	Requester	Own Draft; scope/master/field validation	DRAFT	Audit create/edit; no SLA
ST-RR-002	DRAFT	Submit	Requester	Mandatory/master/scope valid; duplicate rule handled	SUBMITTED	Generate Request No; start SLA; route event
ST-RR-003	SUBMITTED	Route for Review	System	Active approval route / approver	UNDER_REVIEW	Routing audit/ notification; failure stays SUBMITTED
ST-RR-004	SUBMITTED / UNDER_REVIEW	Approve	Approver	Authorized for request scope	APPROVED	Decision audit; notify Coordinator
ST-RR-005	SUBMITTED / UNDER_REVIEW	Reject	Approver	Reason required	REJECTED	SLA stop; breach preserved; notify Requester
ST-RR-006	SUBMITTED / UNDER_REVIEW	Return for Correction	Approver	Reason required	DRAFT	Restore edit permission; decision history retained
ST-RR-007	DRAFT / SUBMITTED / UNDER_REVIEW / APPROVED	Cancel	Requester	Owner; pre-CONVERTED; reason	CANCELLED	SLA stop if started; terminal
ST-RR-008	APPROVED	Create Work Order	Coordinator	No existing WO; current rowversion	CONVERTED	Atomically create WO OPEN; audit/ outbox

3. Work Order Transition Matrix

ID	From	Action	Actor	Guard / Required Data	To	Side Effect / Audit
ST-WO-001	OPEN	Schedule	Coordinator	Active team/site + valid Visit	SCHEDULED	Create Visit SCHEDULED; audit
ST-WO-002	SCHEDULED	Check-in	Assigned Technician	Assigned SCHEDULED Visit; no overlap	IN_PROGRESS	Start Work Session; audit
ST-WO-003	IN_PROGRESS	Submit Work Summary	Team Lead	Session checked out; summary/outcome/evidence complete	AWAITING_SUPERVISOR_REVIEW	Audit + notify Supervisor
ST-WO-004	AWAITING_SUPERVISOR_REVIEW	Submit for Acceptance	Supervisor	Summary reviewed; designated contact valid	AWAITING_CUSTOMER_ACCEPTANCE	Notify designated contact
ST-WO-005	AWAITING_CUSTOMER_ACCEPTANCE	Accept Work	Designated Contact	Active, exact designated contact	COMPLETED	Acceptance round + audit; notify Supervisor/Coordinator
ST-WO-006	COMPLETED	Close	Supervisor	Work Summary + Cost Summary reviewed	CLOSED	SLA stop; lock operations
ST-WO-007	AWAITING_CUSTOMER_ACCEPTANCE	Reject Work	Designated Contact	Reason required	CORRECTIVE_ACTION_REQUIRED	Create CorrectiveAction DRAFT atomically
ST-WO-008	CORRECTIVE_ACTION_REQUIRED	Submit Corrective Plan	Team Lead	Plan + CLEAN evidence complete	CORRECTIVE_PLAN_PENDING	CA -> PENDING_PLAN_APPROVAL
ST-WO-009	CORRECTIVE_PLAN_PENDING	Approve Corrective Plan	Supervisor	Plan complete	CORRECTIVE_PLAN_APPROVED	CA -> APPROVED
ST-WO-010	CORRECTIVE_PLAN_APPROVED	Start Rework	Assigned Technician	Corrective Visit assigned/scheduled	IN_PROGRESS	Corrective session starts
ST-WO-011	OPEN/SCHEDULED/IN_PROGRESS/AWAITING_* / CORRECTIVE_*	Cancel Work Order	Supervisor	Before Customer Accept; reason	CANCELLED	Cancel non-final Visits; keep COMPLETED/MISSED history; SLA stop

4. Service Visit Transition Matrix

ID	From	Action	Actor	Guard	To	Result
ST-SV-001	-	Create Scheduled Visit	Coordinator	WO permits scheduling; valid team/time	SCHEDULED	New initial/follow-up/corrective Visit
ST-SV-002	SCHEDULED	Check-in	Assigned Technician	Assigned; no active overlap	IN_PROGRESS	Open Work Session
ST-SV-003	IN_PROGRESS	Check-out	Assigned Technician	Summary/outcome/evidence + time sequence valid	COMPLETED	Complete Visit/Session
ST-SV-004	SCHEDULED	Reschedule	Coordinator	Reason + new valid window	RESCHEDULED	Audit old/new schedule
ST-SV-005	RESCHEDULED	Confirm Schedule	Coordinator	Valid team/window	SCHEDULED	Ready for visit
ST-SV-006	SCHEDULED	Reassign	Coordinator	Reason + active team/technician	SCHEDULED	Audit old/new assignee
ST-SV-007	SCHEDULED	Cancel Visit	Coordinator	No Check-in; reason	CANCELLED	

ID	From	Action	Actor	Guard	To	Result
						Visit terminal; WO/SLA continue
ST-SV-008	SCHEDULED	Mark Missed	Coordinator	No Check-in; missed reason	MISSED	Original immutable; notify decision owner
ST-SV-009	MISSED	Decide Missed	Coordinator	Decision + reason	MISSED	Original stays MISSED; new SCHEDULED visit only for Reschedule/ Follow-up/Reassign; Cancel creates none

5. Work Session / Corrective Action / Time Correction

ID	From	Action	Actor	Guard	To
ST-WS-001	-	Check-in / create session	Assigned Technician	Visit SCHEDULED; no overlap	CHECKED_IN
ST-WS-002	CHECKED_IN	Pause	Assigned Technician	No active pause	PAUSED
ST-WS-003	PAUSED	Resume	Assigned Technician	Active pause exists	CHECKED_IN
ST-WS-004	CHECKED_IN	Check-out	Assigned Technician	No active pause; required work facts complete	CHECKED_OUT
ST-CA-001	-	Create on Customer Reject	System	Acceptance decision=REJECT	DRAFT
ST-CA-002	DRAFT	Submit Plan	Team Lead	Plan + CLEAN evidence	PENDING_PLAN_APPROVAL
ST-CA-003	PENDING_PLAN_APPROVAL	Approve Plan	Supervisor	Plan valid	APPROVED
ST-TC-001	-	Request Correction	Technician/Team Lead	Supported timestamp + reason	PENDING
ST-TC-002	PENDING	Approve	Supervisor	Sequence valid	APPROVED
ST-TC-003	PENDING	Reject	Supervisor	Reason	REJECTED

6. SLA Events / Notification / Error Control

ID	Trigger	Actor	Rule / Expected Result
EV-SLA-001	Repair Request SUBMITTED	System	started_at=submit; original_due=+24h; effective_due=original_due; risk_at=effective_due-4h
EV-SLA-002	Current UTC >= risk_at, SLA active	System	Emit SLA Risk once per current effective due; no state change
EV-SLA-003	Current UTC > effective_due_at, SLA active	System	Set breached_at once; notify; never clear later
EV-SLA-004	Authorized SLA Override	Supervisor with permission	Reason required; update effective_due; recalc future risk_at; preserve original due/breach
EV-SLA-005	Request REJECTED/CANCELLED or WO CLOSED/CANCELLED	System	Set stopped_at/reason; prior breach stays
ERR-CONC-001	Stale rowversion / duplicate action	System	Reject with controlled conflict; state unchanged; no duplicate transaction

ID	Trigger	Actor	Rule / Expected Result
NTF-FAIL-001	Notification delivery failed	System	Mark FAILED/log/retry; committed business state remains

7. Terminal / Illegal Transition Rules

- REJECTED/CANCELLED Repair Request, CLOSED/CANCELLED Work Order and COMPLETED/CANCELLED/MISSED final Visit history are not reopened in MVP.
- Supervisor cannot Accept/Reject for customer; only designated contact.
- Original MISSED Visit is immutable; D-15 governs follow-up decision.
- Direct database/API status mutation is prohibited; transitions are command actions.

8. Traceability & Baseline Readiness

Transition IDs above are canonical and replace stale ST-* references from previous documents. All downstream Use Case, Data, Test, API and UI documents in this revision use these IDs.

Performance Audit Revision Appendix

ส่วนนี้เป็น revision content ที่มีผลเหนือ technical note เดิมเฉพาะเมื่อมีข้อขัดแย้งกัน โดยไม่เปลี่ยน Business Baseline.

9. Transition Execution - Performance / Concurrency Technical Mapping

Canonical states and transition IDs remain unchanged. This section only states how the existing transition rules are preserved under concurrency and load.

Transition Class	Technical Execution Constraint	Verification
State-changing commands	Authenticate/authorize -> validate current state/guard/rowversion -> execute one atomic transaction -> append audit -> insert outbox/recipients -> commit -> return fresh ETag.	Integration + TC-SEC-002 + TC-PERF-006
Stale/parallel action	A stale token or duplicate attempt returns controlled conflict; state is unchanged and no duplicate WO/acceptance/notification is created.	ERR-CONC-001 + concurrency load evidence
SLA Risk/Breach events	Worker detection must be idempotent; technical polling/batch changes must not move T-4h, 24h, sticky-breach or stop semantics.	EV-SLA-002/003 + TC-PERF-004
Notification failure	Delivery/retry stays outside the committed business transaction and cannot reverse the transition.	NTF-FAIL-001 / TC-NTF-001
MISSED / Corrective	Performance optimization may not coalesce, skip or reopen immutable/final history.	ST-SV-008/009; ST-CA-* tests

10. Locking / Throughput Boundary

The State document does not prescribe SQL lock hints, isolation-level tuning, batch size or worker degree of parallelism. Those are Technical Design decisions verified by Database/Architecture/Performance tests. Any such choice must retain atomicity and the canonical transition outcome.